

Phase-Aware Reinforcement Learning Based Page Replacement Policy for Adaptive Memory Management

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Traditional page replacement policies such as LRU, CLOCK, and LFU rely on static heuristics like recency or frequency, which perform sub optimally under dynamic workloads exhibiting distinct phases—such as sequential streaming followed by random access loops. These policies do not adapt to phase transitions, leading to elevated page fault rates and increased average memory access times, particularly in modern applications such as databases, ML training, and web servers.

This project proposes a novel Reinforcement Learning (RL)-based page replacement algorithm that dynamically adapts to workload phases. The RL agent observes state features including page recency, frequency, reuse distance approximation, and access pattern locality, then selects eviction actions to maximize long-term rewards defined as minimized miss rates and access latency. Unlike fixed heuristic policies, our phase-aware approach employs an online Q-learning mechanism to detect and respond to workload shifts in seconds, achieving near-optimal performance without prior trace knowledge.

We will implement a high-fidelity page cache simulator in C++/Python, integrating a lightweight RL agent using tabular Q-learning or a small neural network. The algorithm will be rigorously evaluated using synthetic traces and real-world workloads (example SPEC, YCSB), measuring key metrics: page miss rate, hit rate, and average memory access time across varying cache sizes (64–4096 pages). Comparisons against LRU, CLOCK, LFU and Belady’s offline optimal algorithm will demonstrate statistically significant improvements (aiming to reduce 10–25 miss rate) in mixed-phase scenarios, proving superior adaptability and efficiency.

References

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